

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js automation service** for MCA filings, reducing manual filing time by **85%** with retries, browser-state recovery, and API-backed validation.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized backend services with **Docker**, deployed to **Google Cloud Run**, and automated CI/CD with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, integrating model inference, preprocessing, and visualization into an existing desktop medical-imaging application.
- Reduced processing time by **87%** using asynchronous and parallel execution across expensive image-processing workflows.
- Improved 3D reconstructed surface visualization while preserving compatibility with existing VTK/wxPython user flows.

Open Source

[Contributions](#)

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, adapting implementation details to established open-source review and release practices.

Projects

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#)

- Built containerized Go/Rod services for a **content distribution platform** using browser automation, with async workers, RabbitMQ retries, Redis state, and PostgreSQL persistence.
- Designed the system for operational simplicity, service isolation, and recoverable execution around unreliable external integrations and long-running jobs.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#)

- Built a self-hosted **AI security gateway** for LLM, HTTP tool, and MCP traffic, blocking **90%+** of prompt-injection and unsafe-tool attempts across a **150-case attack corpus** with false positives under **5%**.
- Added route profiles, RBAC runtime settings, PII redaction, allow/block rules, **OpenTelemetry** metrics/traces, health checks, and audit ledgers for **100%** of policy decisions.

Distributed Inference Engine | *Python, PyTorch, FastAPI, gRPC, CUDA, Docker, Redis*

- Built a **distributed inference engine** to run models larger than a single consumer GPU by coordinating token generation across heterogeneous nodes with speculative decoding.
- Designed streaming APIs, worker heartbeats, retry-aware orchestration, and KV-cache-aware routing to balance latency, throughput, and worker availability across local and remote inference workers.

Technical Skills

Languages: Go, Python, TypeScript

Frameworks: Echo, FastAPI, Gin, Node.js

Tools: Docker, Google Cloud Run, GitHub Actions, PostgreSQL, Redis, RabbitMQ, Prometheus, AWS